TEAM THERMOCLINE

snhu

Our Project

We are a team of engineering students at Southern New Hampshire University completing our capstone project. We are one of two teams making a thermal testing chamber this year!

Purpose of a Thermal Testing Chamber

In industry, thermal testing chambers are used to verify the operation of electronics and devices at various temperature ranges.

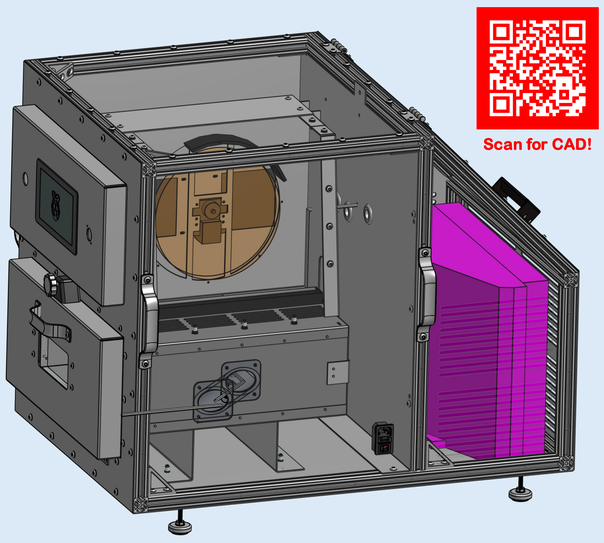
Photo Credit LIBtest.com

Testing chambers aid in verifying individual components and products function in many environments. They can also be useful for specific heat and thermal testing in other areas, such as unique materials, freeze-drying, and many applications.

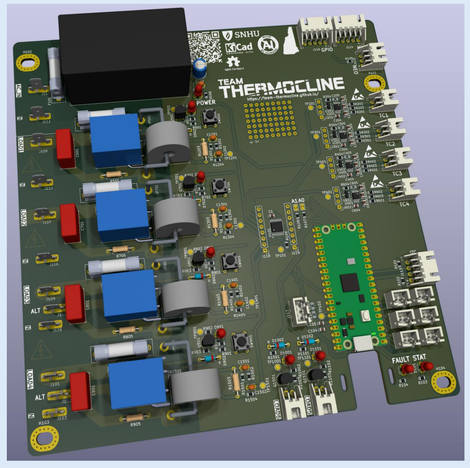
Our Design Process

The team uses tools such as *Autodesk Inventor*, *FreeCAD* and *Onshape* to collaborate and design the chamber in 3D. The testing chamber is made from stamped and bent laser cut sheet metal parts and aluminum extrusion. The chamber is assembled with rivets, bolts and welds.

The body of the chamber keeps the user safe from extreme temperatures, hazardous insulation, and refrigerant.



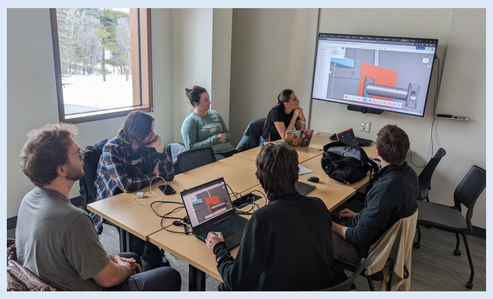
Scan for CAD!



KiCad was used to make the mainboard PCB and other test or adapter PCBs.

Scan for schematics!

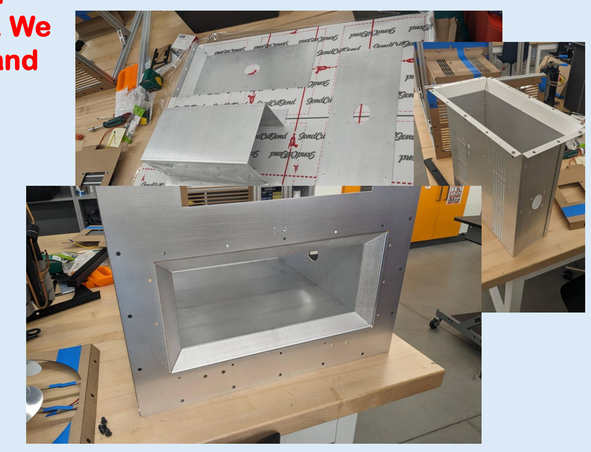
The team had both mechanical and electrical review processes, where we asked educators and industry engineers for input on our designs before sending them off to be made!

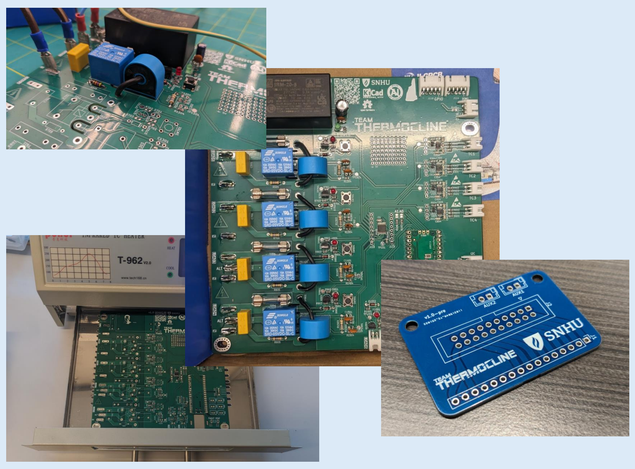


Fabrication

We used SendCutSend for the manufacturing of the sheet metal parts. Parts arrived laser cut and bent to our CAD specifications. We assembled them after arrival using rivets, RTV silicone, rivnuts, and other fasteners and gaskets.

SendCutSend



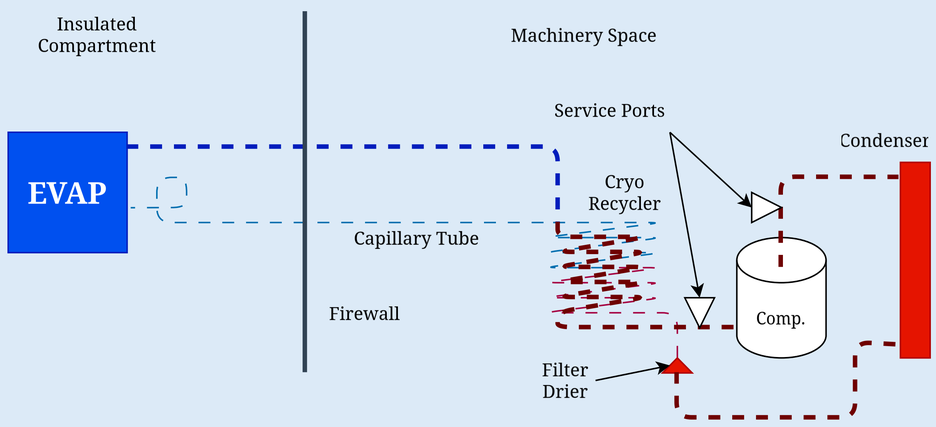


JLCPCB was used as the manufacturer for the printed circuit boards, we rendered manufacturing files and had them shipped out. Shipping from China can take a while, so we always factored in lead times when ordering parts. Blank boards arrive in the mail and are assembled at SNHU by hand. We used LCSC as a parts supplier.

JLCPCB

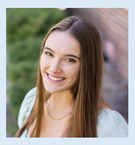
Refrigeration System

Charged with 1lb of R410A, the system is around 2000cc internal volume. The absolute boiling point of R410A is –43C (-45F) at vacuum. We exploit this property by using 10ft of capillary tubing to restrict the flow much more than a normal refrigeration system.



Category	Type	Count
Frame	Aluminum Extrusion	28ft
	Aluminum Panels	25ct
Fasteners	Bolts	>250ct
	Rivet Nuts	100ct
	Pop Rivets	50ct
Thermal	Copper	10lb
Denim Insulation	Denim Insulation	$5ft^3$

About the Team



Alexandra Friebolin

Mechanical Engineer

Jacob Morrissette

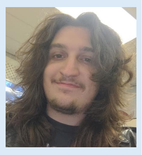
Mechanical Engineer

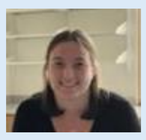
Nik DiLullo

Electrical Engineer

Joe Sedutto

Electrical Engineer





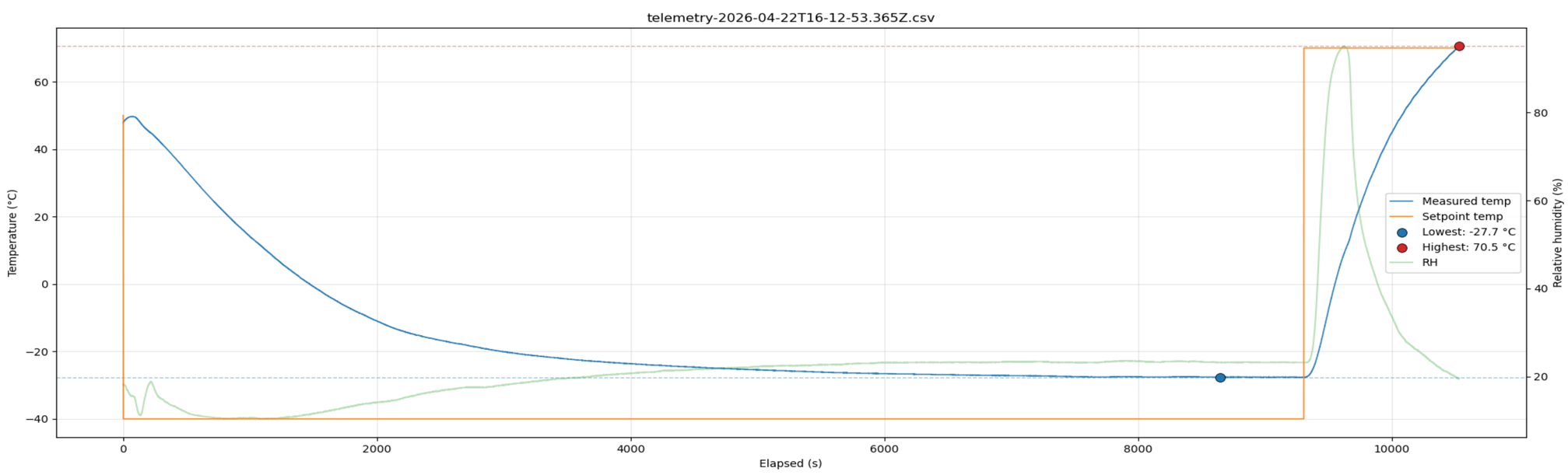
Alexia Hnatowicz

Mechanical Engineer

Final Results

With evaporator temperatures nearing –36C (-32F) we've been able to meet all functional requirements related to temperature, ramp-rate and cycle time! With our current best minimum sustained air temperature at –27C (-17F).

Specification	Result
Minimum Air Temperature	-27C (1.5F)
Maximum Air Temperature	75C (167F)
Maximum Cooling Ramp Rate	-1.5C/min
Maximum Heating Ramp Rate	>5C/min



View our CAD, documentation, design process and more on our website!